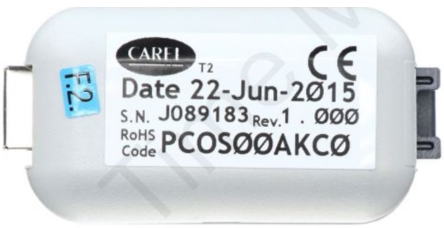
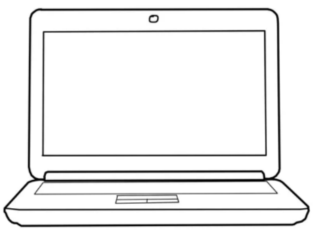
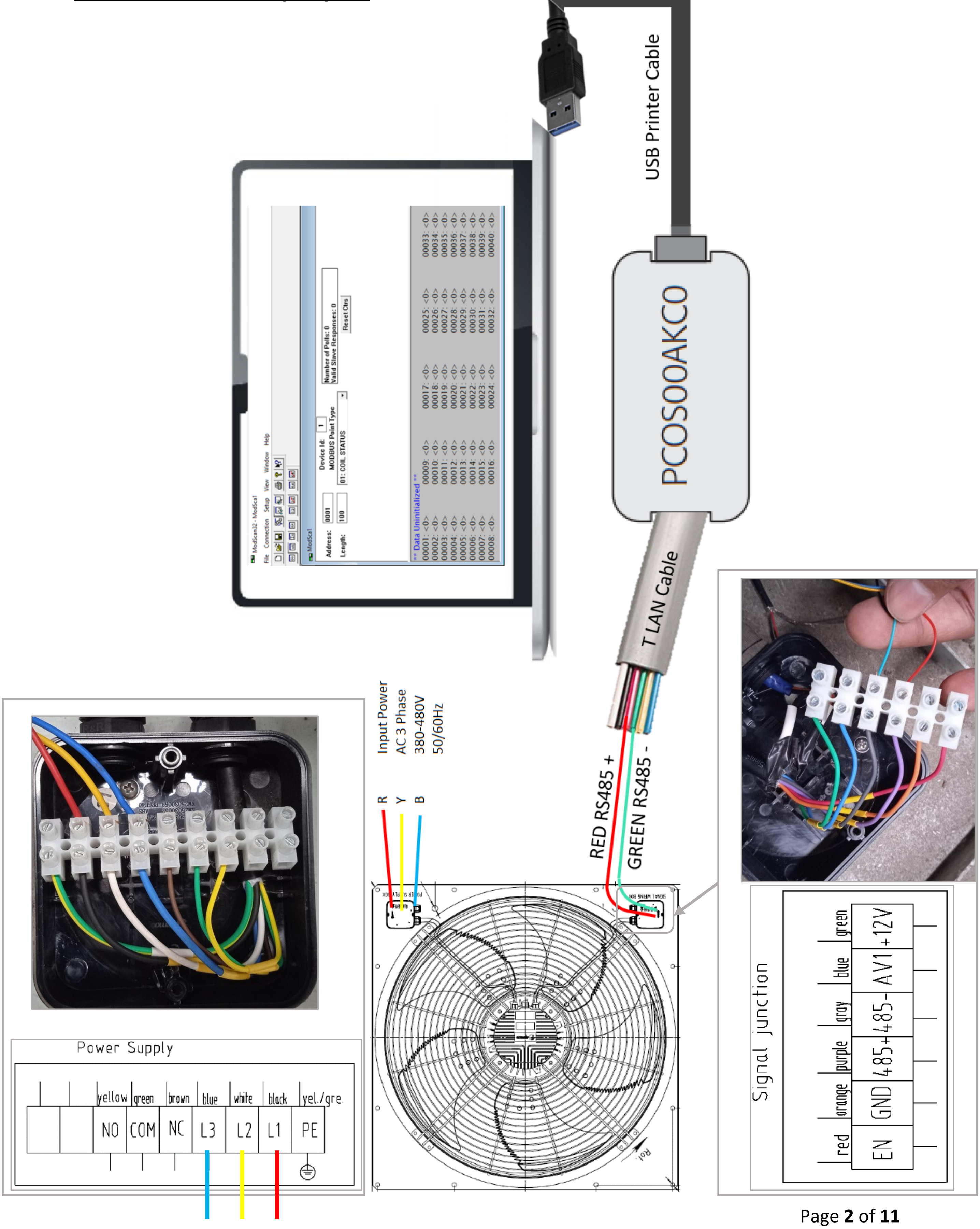
Scope: This document will help the user to established a RS485 to USB communication network and configure it’s parameter as per KCPL requirement in EC Axial Fan Model No - ODS950C-120B5.DC.V-KCPL manufactured by Kemao and used in Air cooled chillers.

Required Tools: EC Axial Fan, Modscan32 software installed Laptop, Carel PCOS00AKC0 Smart Key USB adaptor, Printer USB cable, Telephone-LAN Cable, 380-480V, 50/60Hz, 3 phase input power supply.

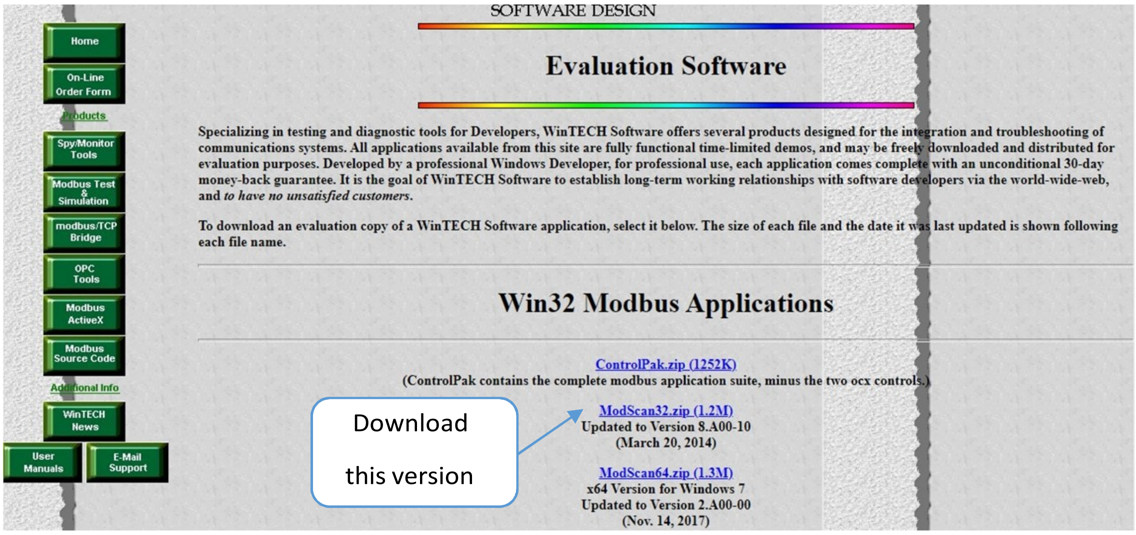




Power and Control Wiring Daigram:



<u>MOD Scan Software Installation:</u> Open any search engine and go to this link--modscan32.zip



Download MoDscan32.zip and Extract this file

After extraction the MoDscan Zip file have a setup file (Setup file doesn't need to installation in PC). Open setup and ignore all the login section by the cancellation and its ready for testing.

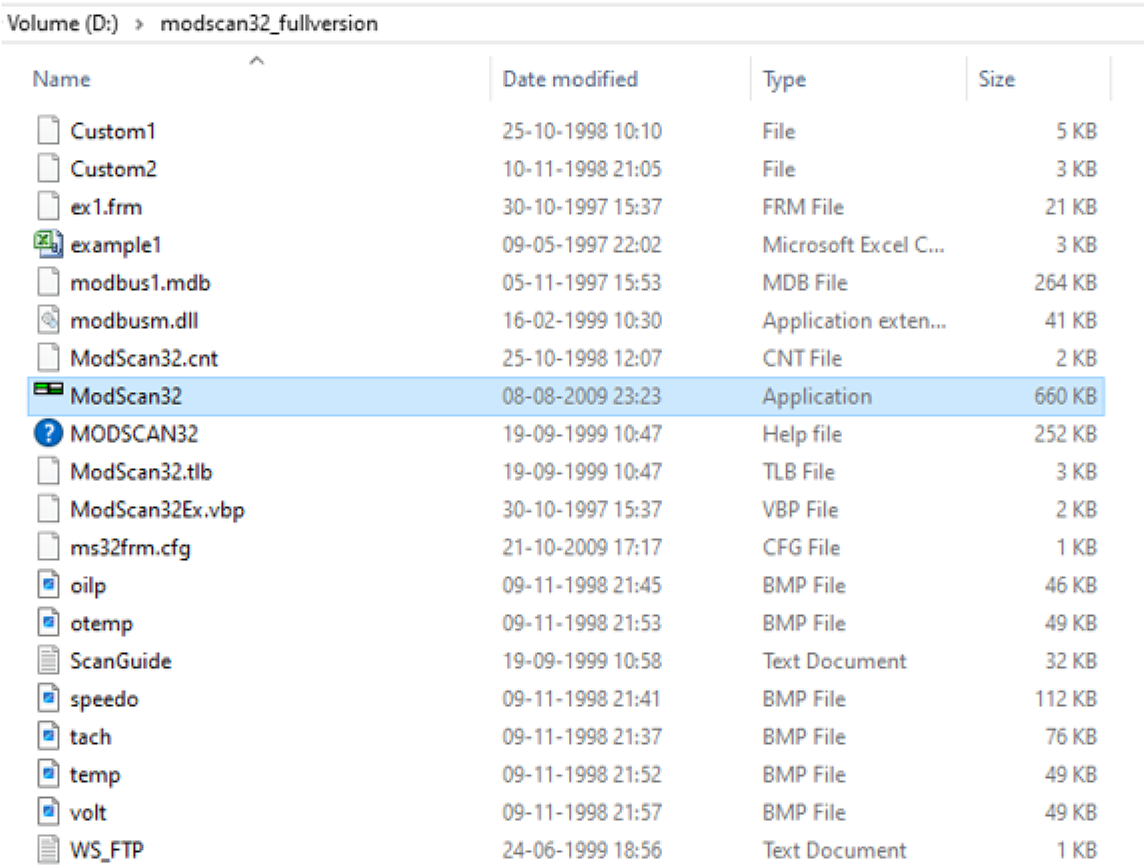
<u>Carel PCOS00AKC0 Smart Key Driver Installation:</u>

The operation of the adapter requires the installation of the VCP driver, the same used by the USB-RS485 converter, code CVSTDUTLF0 or CVSTDUMOR0. Consequently, if one of these devices is already used no new driver needs to be installed. To get the VCP driver (Virtual Com Port), simply go to the CAREL site http://ksa.carel.com, under the free download area (USB_RS-485_CAREL_Converter section) or download the fi le corresponding to the FT232BM component directly from http://www.ftdichip.com/FTDrivers.htm. When connecting the adapter for the first time to a USB port on the personal computer, the system detects the presence of new hardware and requests the installation the VCP driver for the management of the serial port; at this point select the path of the directory where the drivers were previously saved and decompressed. If the drivers were installed correctly, the system automatically detects a new COM port, usually COM3. If more than one adapter is connected to the same PC, check under My Computer + right button (mouse) > Properties > Hardware > Peripherals > Ports (COM and LPT) which port has been assigned by the system to the PCOS00AKC0 adapter.

<u>Starting Serial Communication with Fan:</u>

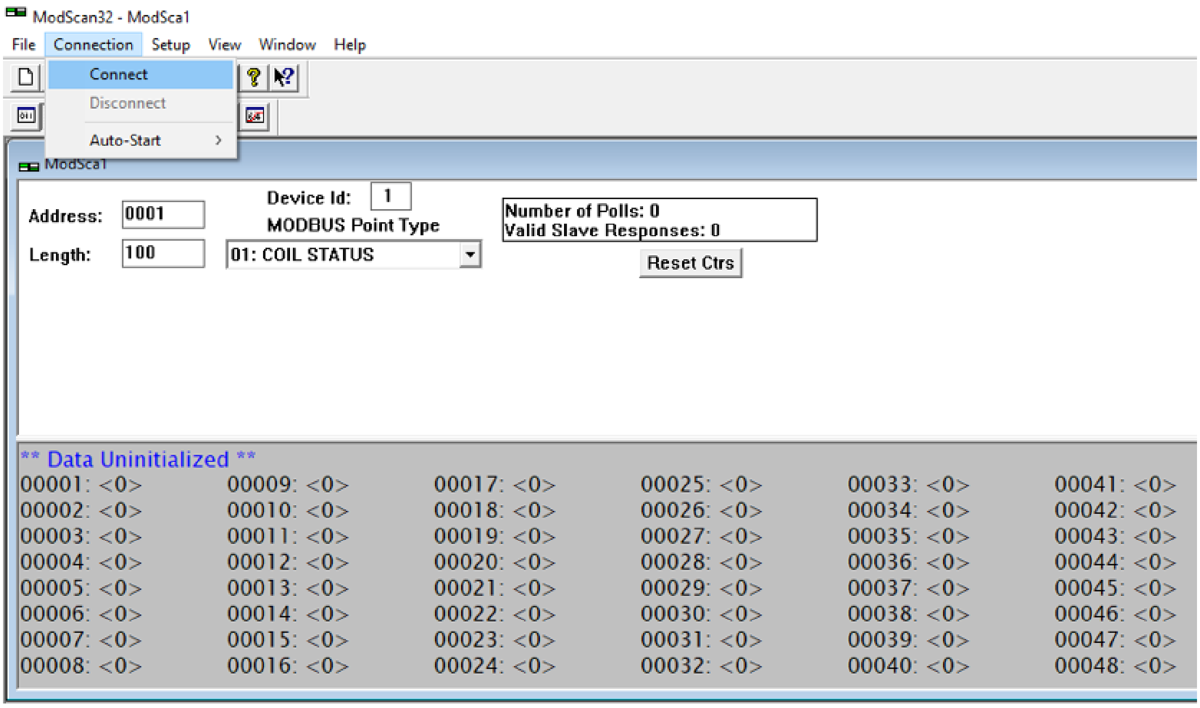
1. Check Power and Control wiring connection as per diagram.
2. Switch on 3 phase input power supply to Fan.
3. Plug in the USB cable to Laptop and observe the communication of Carel PCOS00AKC0 Smart Key USB adaptor with Laptop.

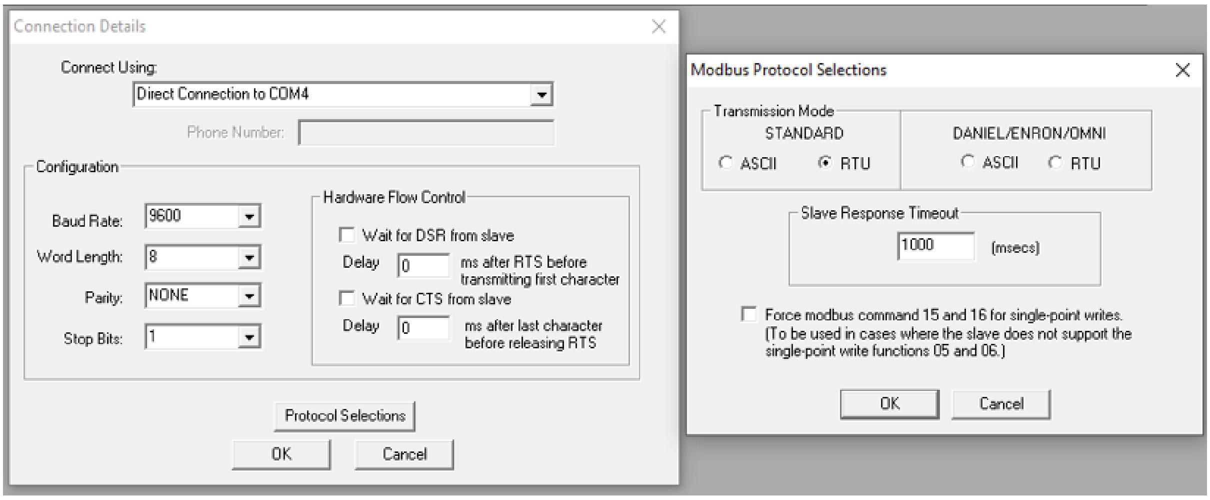
4. Run the ModScan32 Application file in laptop by double click of mouse button.


Volume (D:) › modscan32_fullversion

Name	Date modified	Type	Size
Custom1	25-10-1998 10:10	File	5 KB
Custom2	10-11-1998 21:05	File	3 KB
ex1.frm	30-10-1997 15:37	FRM File	21 KB
example1	09-05-1997 22:02	Microsoft Excel C...	3 KB
modbus1.mdb	05-11-1997 15:53	MDB File	264 KB
modbusm.dll	16-02-1999 10:30	Application exten...	41 KB
ModScan32.cnt	25-10-1998 12:07	CNT File	2 KB
ModScan32	08-08-2009 23:23	Application	660 KB
MODSCAN32	19-09-1999 10:47	Help file	252 KB
ModScan32.tlb	19-09-1999 10:47	TLB File	3 KB
ModScan32Ex.vbp	30-10-1997 15:37	VBP File	2 KB
ms32frm.cfg	21-10-2009 17:17	CFG File	1 KB
oilp	09-11-1998 21:45	BMP File	46 KB
otemp	09-11-1998 21:53	BMP File	49 KB
ScanGuide	19-09-1999 10:58	Text Document	32 KB
speedo	09-11-1998 21:41	BMP File	112 KB
tach	09-11-1998 21:37	BMP File	76 KB
temp	09-11-1998 21:52	BMP File	49 KB
volt	09-11-1998 21:57	BMP File	49 KB
WS_FTP	24-06-1999 18:56	Text Document	1 KB

5. Click on Connect and confirm the the connection detail as per below screen short.



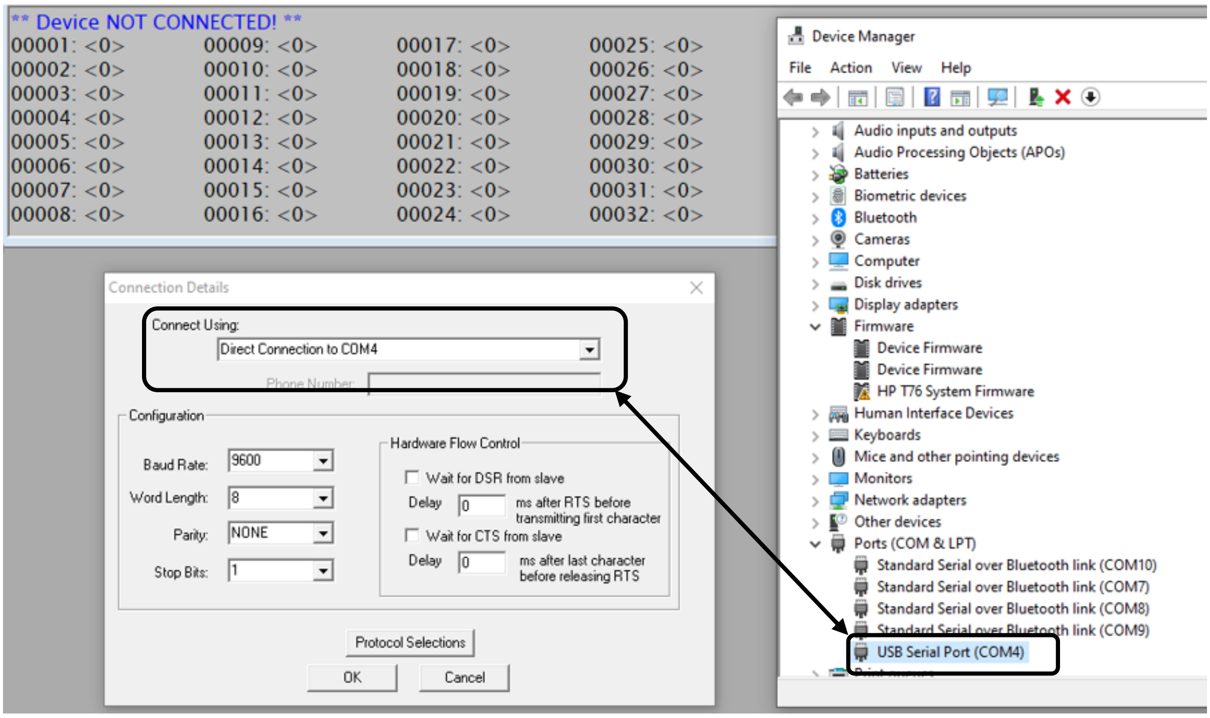


We receive following default communication setting which is done by Fan manufacturer.

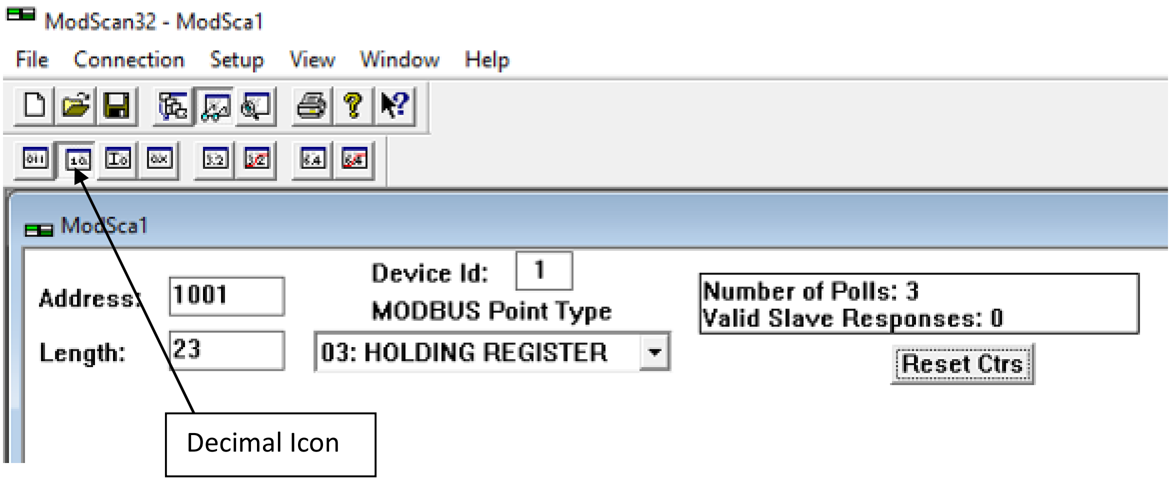
Baud Rate : 9600
Word Length : 8
Parity : None
Stop Bit : 1
Modbus Device ID : 1

Note - During first time communication at KCPL factory this setting works fine, for next time and onwards Baud rate need to change to 192000.

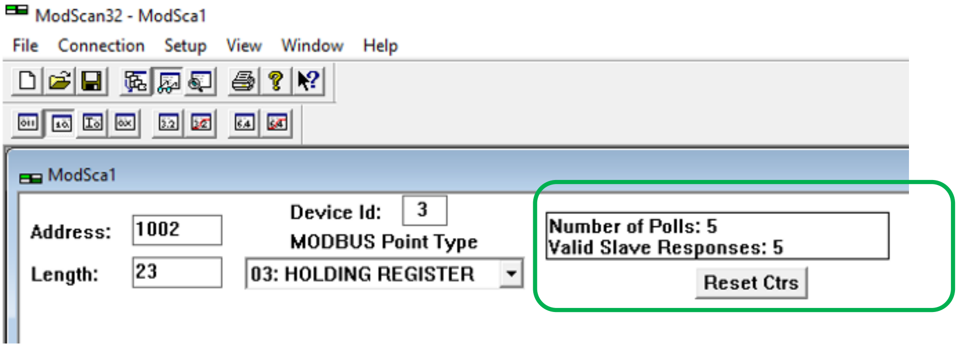
6. Check the COM port selection in drop down box under heading “Connect using” in this case COM4 USB port is assigned to ModScan32 software for Modbus communication with Fan. Carel PCOS00AKC0 Smart Key USB adaptor must be plugged in this USB port only.



7. In ModScan Software screen click on **Decimal** icon tab and fill the input data box as per below image.



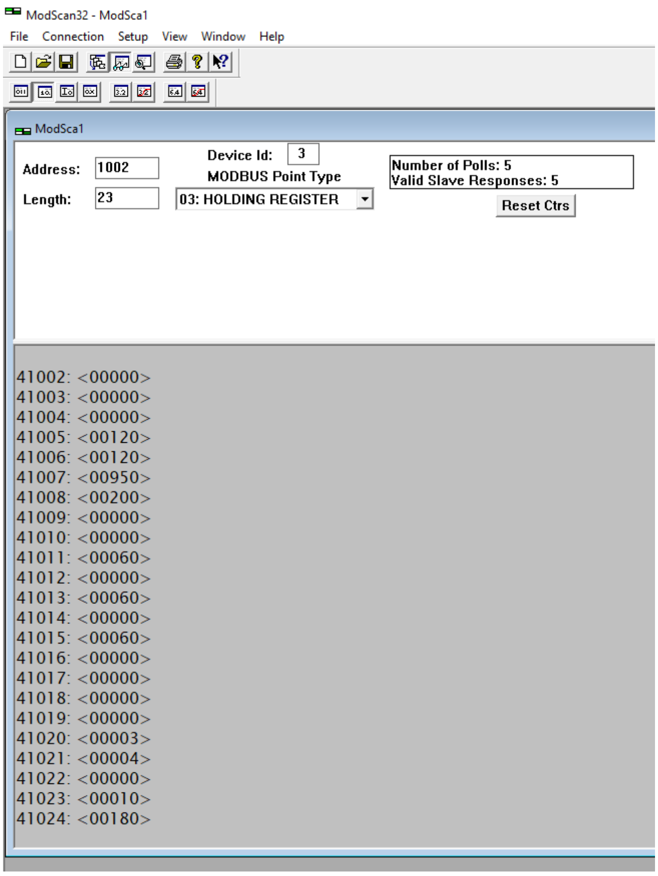
8. Serial communication with fan will be start, see the increasing counter of Valid Slave Response marked in green box of below image.

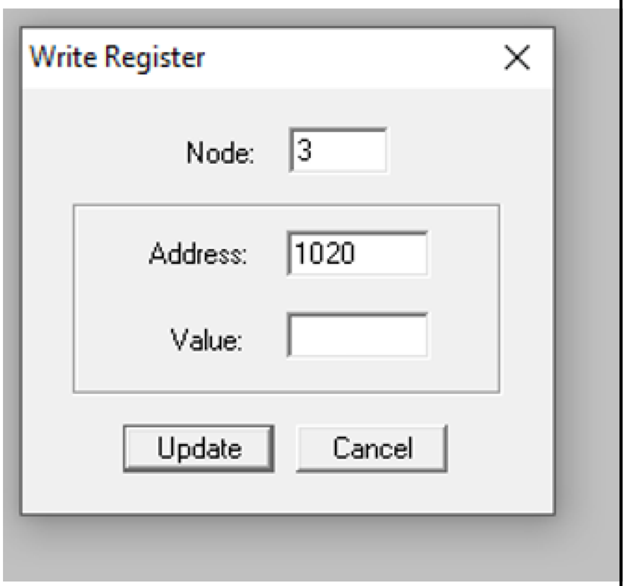


9. Double click to change the Modbus Holding Register 41020 from default value 1 to any suitable number (1 to 20) as per installation location of the fan specified in last page. In this step, we have assign a Device Id (Modbus address number to the connected fan).

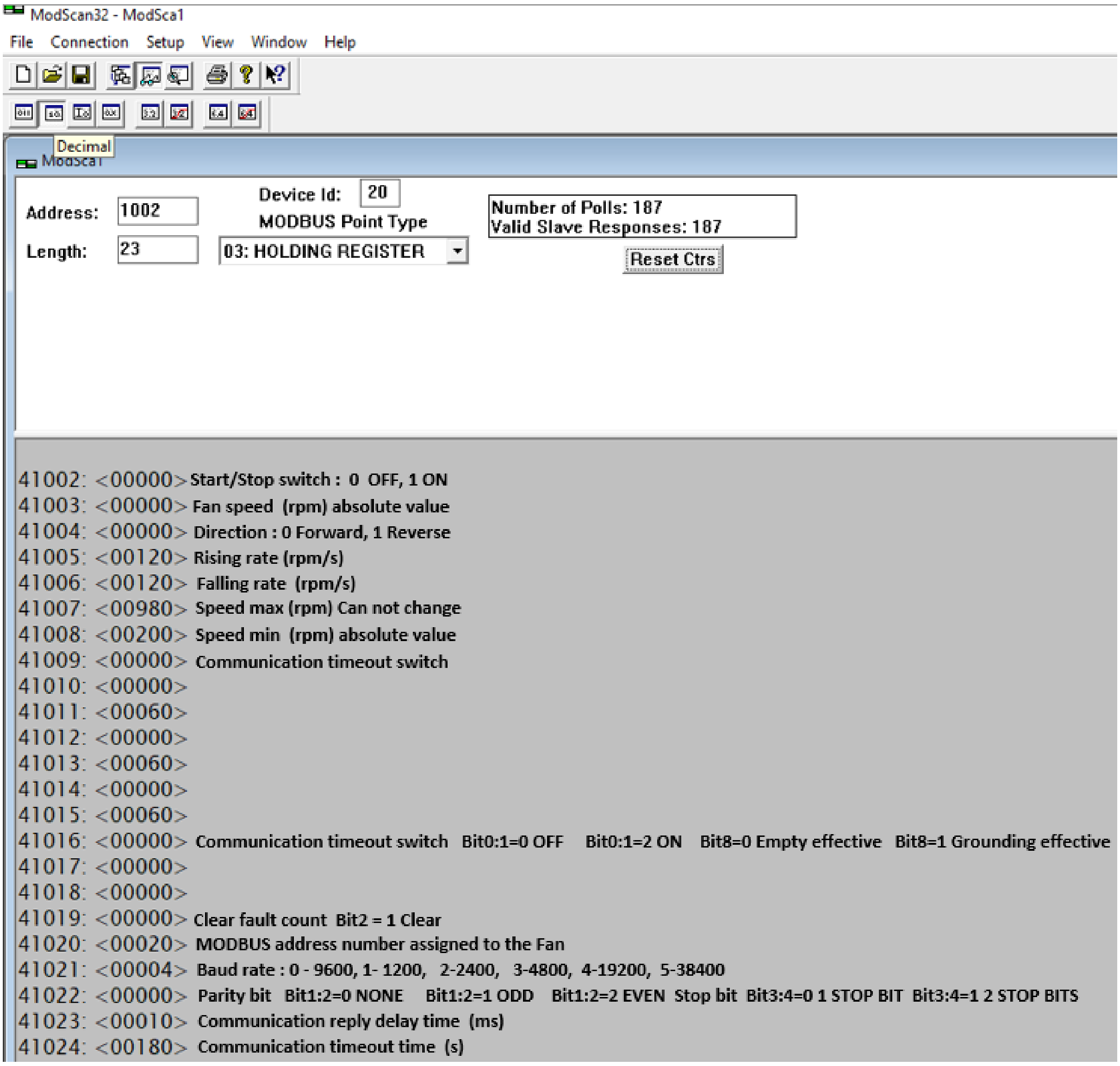
Caution !

Re-confirm and note down the type value editing for Modbus Holding Register 41020 and (take a screenshot also) before click on Update button.
This number has significant role in re-establishing a communication with the fan.
Failed to do so Modbus communication will be lost and not possible to re-connect.
Defective fan module need to send back to its manufacturer for reset.





10. Disconnect the input power supply for at least 180 second and then re-connect. Write value for the remaining register as per below and verify it.

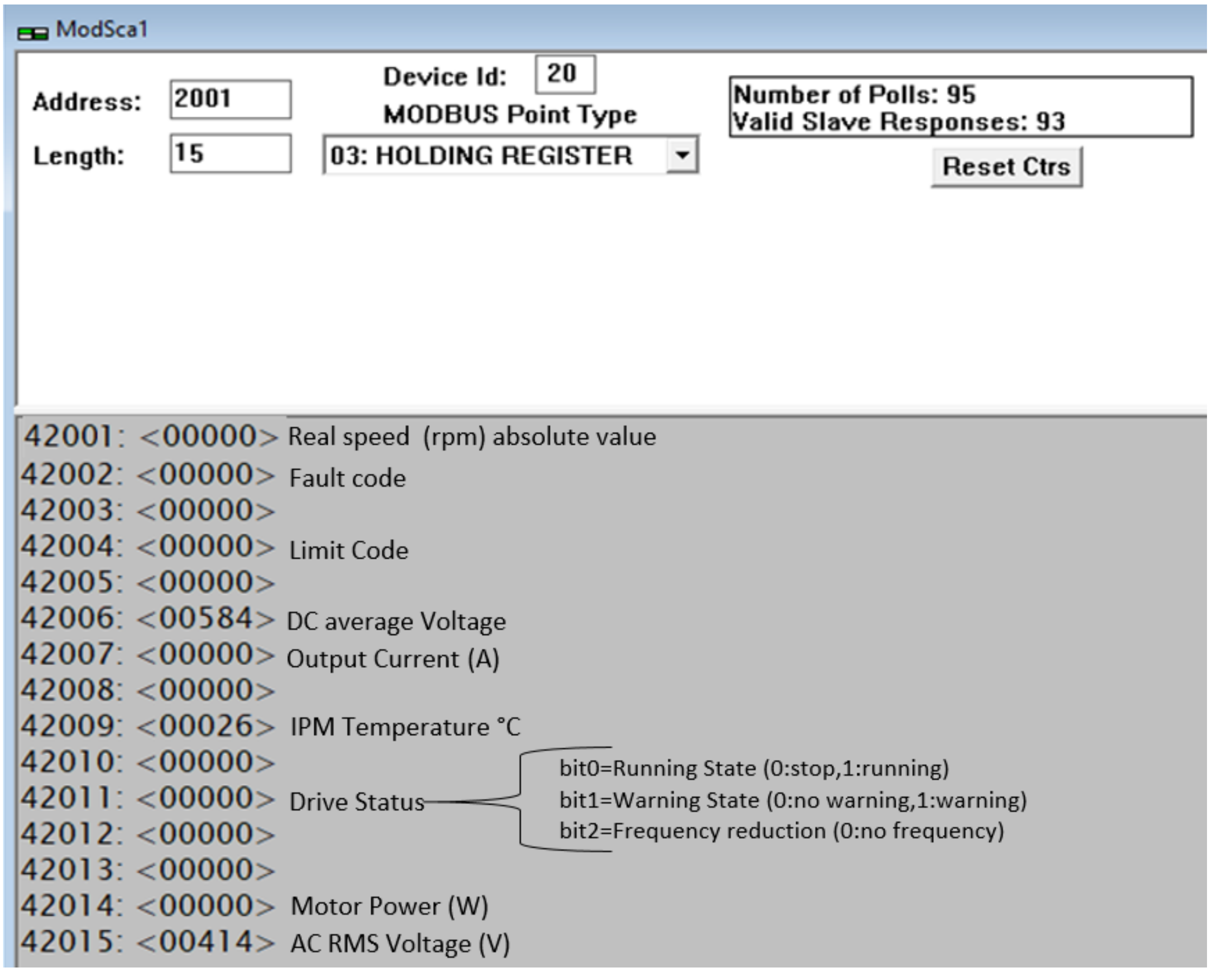


Note - If the outlet air temperature of the condenser coil is more than 60°C, keep Maximum speed setting to 950 rpm to avoid un-necessary tripping of the fan.

11. Disconnect the power supply for at least 180 second and re-connect. Stablish communication with ModScan32 software with Baud rate 19200 and check all set point value written in previous stage. There must not be any deviation. If so, repeat the above process.

12. Now, Fan is ready for rotation check test. Write the Register Address 41003 value to 500, then write the Register Address 41002 value to 1. Allow the fan to run for sufficient time and observe for any abnormal noise or any other defect.

13. Open the Monitoring screen to read the important parameters.

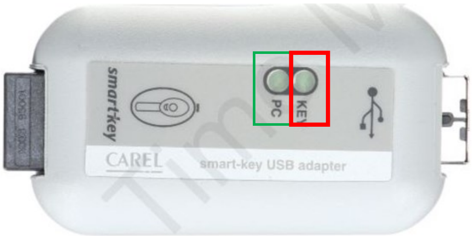


14. If Fan is performing well then to stop it, first change the register address 41002 value to 0 then also change the register address 41003 value to 0.

15. Configuration process is completed now. Disconnect the input power supply, remove the communication wire. It can be use in assembly of Chiller product.

Troubleshooting for communication problem

1. If you do not see the increase number in Valid Slave Response counter, check Number of Polls counter and Smart key LED status.



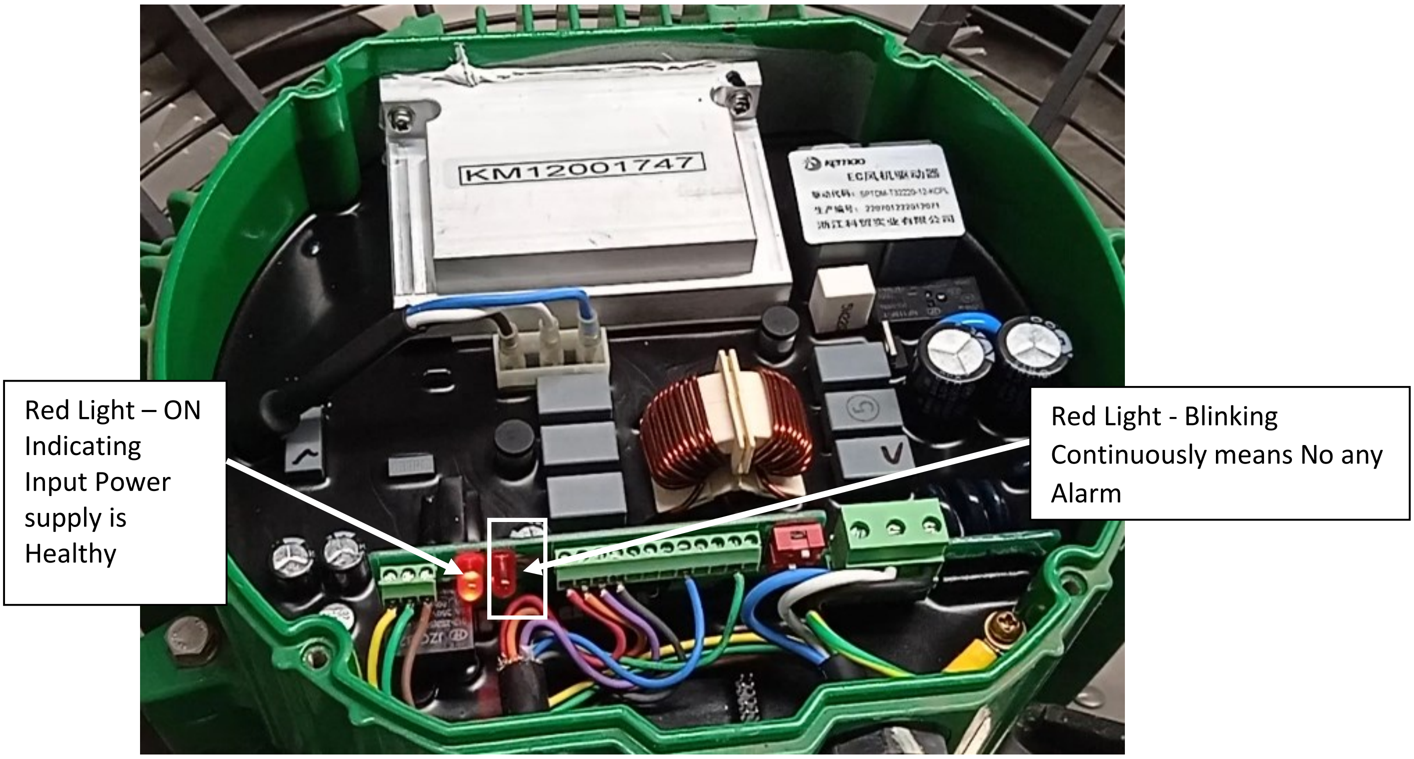
If PC light (marked in Green box) is blinking and Key Light (marked in Red box) is continuously OFF, this is indication of something is wrong in between Smart key and Fan.

Re-check input power supply status to the Fan.

Try with a different Baud rate which is 192000, If still no response is received try to communicate with Fan with a different Modbus Device ID (1 to 20).

2. At last, disconnect input power supply and open the end cover of Fan assembly to check the status of power circuit board.

Once cover plate is open, switch on the power supply and monitor the both LED status



3. If input Power supply is OK and there is no any alarm, then re-check control wiring

Take support from Fan Manufacturer for further troubleshooting.

Electric Panel			
Fan No - 2 Modbus address -2			Fan No - 1 Modbus address -1
Fan No - 3 Modbus address -6			Fan No - 4 Modbus address -5
Fan No - 5 Modbus address -10			Fan No - 6 Modbus address -9
Fan No - 7 Modbus address -14			Fan No - 8 Modbus address -13
Fan No - 9 Modbus address -18			Fan No - 10 Modbus address -17
Fan No - 2 Modbus address -20	**Not applicable for 18 Fan chiller model**		Fan No - 1 Modbus address -19
Fan No - 3 Modbus address -16			Fan No - 4 Modbus address - 15
Fan No - 5 Modbus address -12			Fan No - 6 Modbus address -11
Fan No - 7 Modbus address -8			Fan No - 8 Modbus address -7
Fan No - 9 Modbus address - 4			Fan No - 10 Modbus address -3